



Classes 3 and 4 — Network Access Control and Security

Cyber Defense (420-D03-AB) — Lecture for the students — Cyber.SoHo

Class 3 (Wednesday, September 2) and Class 4 (Tuesday, September 8) — LAB_02 introduction — delivered Tuesday, September 8, 2026

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Read this first — the rules that apply to everything today

- All material is educational only: it shows how to protect systems, never how to attack the systems of others.
- You practice only inside your personal, isolated laboratory, or on a system with explicit written permission obtained before you start.
- Unauthorized computer access is a criminal offence — Criminal Code of Canada, sections 342.1, 342.2 and 430(1.1).
- You sign the Acknowledgment and Agreement page at the beginning of each session.
- **AI** (Artificial Intelligence) tools are allowed in every submission, provided the chapter states the **LLM** (Large Language Model) used and the exact prompts, verbatim.
- Laboratory-only accounts and passwords: nothing real ever appears in a screenshot.



Today in 90 minutes

Segment	Minutes
Opening — session note, session signature, agenda	5
1.1 Privilege and access management	10
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Where today's work goes

- LAB_02 — Privilege and Access Management (5%): users, groups, least-privilege memberships, folder permissions, test logins.
- Chapter 2 of the Final Project and of the Course Procedure Manual: the access-control chapter.
- LAB_03 (Class 5–6) reuses today's account policies and sudo rule in the hardening checklist.
- Chapter 7 — resilience assessment: rows 1, 2, 8, 9 and 10 are answered by today's layers.
- Every script shown today is an accepted starting point for your own work, with disclosure in the manual.



Lesson 1 — Class 3

Network Access Control and Security — part 1

- 1.1 Privilege and access management
- 1.2 Least privilege and need-to-know
- 1.3 Users, groups and roles on Windows and Linux
- 1.4 Granting, reviewing and removing access



1.1 Access control: subject, operation, object (1/2)

- Access control decides whether a subject may perform an operation on an object.
- Subject: a user account, a service, a program acting for a user.
- Operation: read, write, execute, delete, change permissions.
- Object: a file, a folder, a database, a network device, a service.
- Identity → Authentication → Authorization → Accounting: the **AAA** (Authentication, Authorization, Accounting) model.
- Permission = a right on an object. Privilege (user right) = a right of an identity on the whole system.



1.1 Models and the access matrix (2/2)

- **DAC** (Discretionary Access Control): the owner of the object decides (**NTFS** (New Technology File System), Linux modes).
- **MAC** (Mandatory Access Control): a central policy and labels decide (SELinux, AppArmor) — awareness.
- **RBAC** (Role-Based Access Control): permissions on roles, roles as groups — the model of every lab.
- Access matrix: rows = roles, columns = systems, cells = NONE / READ / USE / ADMIN.
- Rule checked today: only the **IT** (Information Technology) administrator holds ADMIN in the server zone.

Four privilege levels

- NONE — no account, no access
- READ — look only
- USE — read and write the data of the role
- ADMIN — change the system itself



1.1 The access matrix of NorthMaple Accounting Inc.

Access matrix of NorthMaple Accounting Inc. — who may do what, on which system						
Role \ System	Firewall	Web server (DMZ)	File server	Windows workstation	Linux workstation	VPN gateway
Administrator (IT)	ADMIN	ADMIN	ADMIN	ADMIN	ADMIN	ADMIN
Accountant	NONE	NONE	USE	USE	USE	NONE
Technician	NONE	NONE	USE	ADMIN	ADMIN	NONE
Receptionist	NONE	NONE	USE	USE	NONE	NONE
Remote worker	NONE	NONE	USE	NONE	NONE	USE

Server zone and perimeter: ADMIN belongs to the IT administrator only (policy check of nm_access_matrix.py: PASS).

Privilege levels:

NONE	no account, no access	READ	look only
USE	read and write the data of the role	ADMIN	change the system itself

The remote worker is an accountant working from home: USE on the file server, reachable only through the VPN gateway (LAB_06). Cyber.SoHo — Cyber Defense (420-D03-AB) — Classes 3 and 4

One row per role, one column per system; ADMIN in the server zone belongs to one role only.



1.1 Example and script — nm_access_matrix.py

- Task: build the matrix from the role table, print it, check the server-zone rule, save it as **CSV** (Comma-Separated Values).
- Runner buttons in the lecture document: Copy, Start / Pause, Step, Restart, Download.
- Pause after the header row: predict the Technician row before resuming.
- Policy check: PASS when no user role holds ADMIN on a server-zone system.
- Deliverable: northmaple_access_matrix.csv, attached to Chapter 2 of the manual.



1.2 Least privilege (1/2)

- Every identity holds only the permissions and privileges strictly required for its task, for the shortest time — nothing more (**NIST** (National Institute of Standards and Technology) SP 800-53, AC-6).
- Daily work never uses an administrative account; administrators hold two accounts.
- Permissions go to the role (the group), never to the person.
- Rights no longer needed are removed — otherwise privilege creep accumulates.
- A program never obtains more rights than the account that started it (unless it exploits a vulnerability).



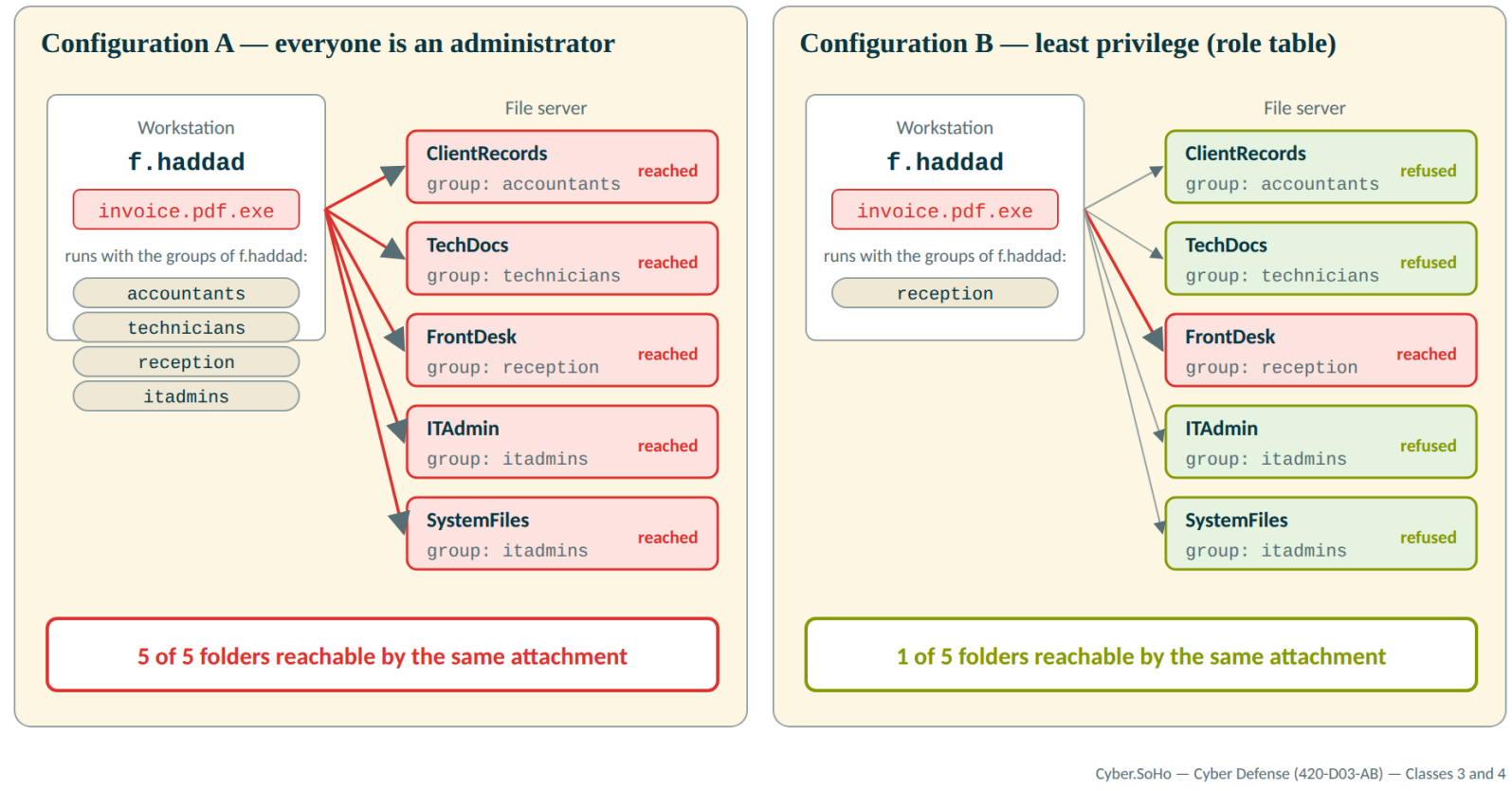
1.2 Need-to-know and separation of duties (2/2)

- Need-to-know: access to a piece of information only if the current task requires it — even inside an allowed privilege level.
- Least privilege limits the level; need-to-know limits the scope.
- Data classification at NorthMaple: public, internal, confidential.
- Separation of duties: no single identity completes a sensitive operation alone (awareness).
- Physical access control follows the same design: one key per role that needs it.
- Apply in order: start at NONE, add the minimum, use groups, elevate for the task only, review.



1.2 Reach of one infected account

Reach of one infected account: everyone-is-admin versus least privilege



The same attachment, opened by the same receptionist account: five folders exposed, or one.

1.2 Example — nm_least_privilege_reach.py

5 / 5 → 1 / 5

- Folders reachable by a program started by f.haddad: five out of five when everyone is an administrator, one out of five with the role table.
- The script only compares group lists — it never runs a program.
- The two counts are the justification sentence of the least-privilege memberships in Chapter 2.



1.3 Identities on Windows and Linux (1/2)

- Identity = a record: name, unique identifier, home folder or profile, shell or logon method, attributes.
- Windows: local accounts in the **SAM** (Security Account Manager); domain accounts in Active Directory; every account and group has a **SID** (Security Identifier).
- Permissions store SIDs, not names: renaming changes nothing; delete and re-create gives a new **SID**.
- Linux: /etc/passwd (name, **UID** (User Identifier), primary **GID** (Group Identifier), home, shell), /etc/shadow (hashes, root only), /etc/group (members).
- root is **UID 0**; the administrative group is sudo (Debian, Ubuntu) or wheel (Fedora, Red Hat).

Built-in Windows groups

- Administrators
- Users
- Guests
- Remote Desktop Users
- Backup Operators

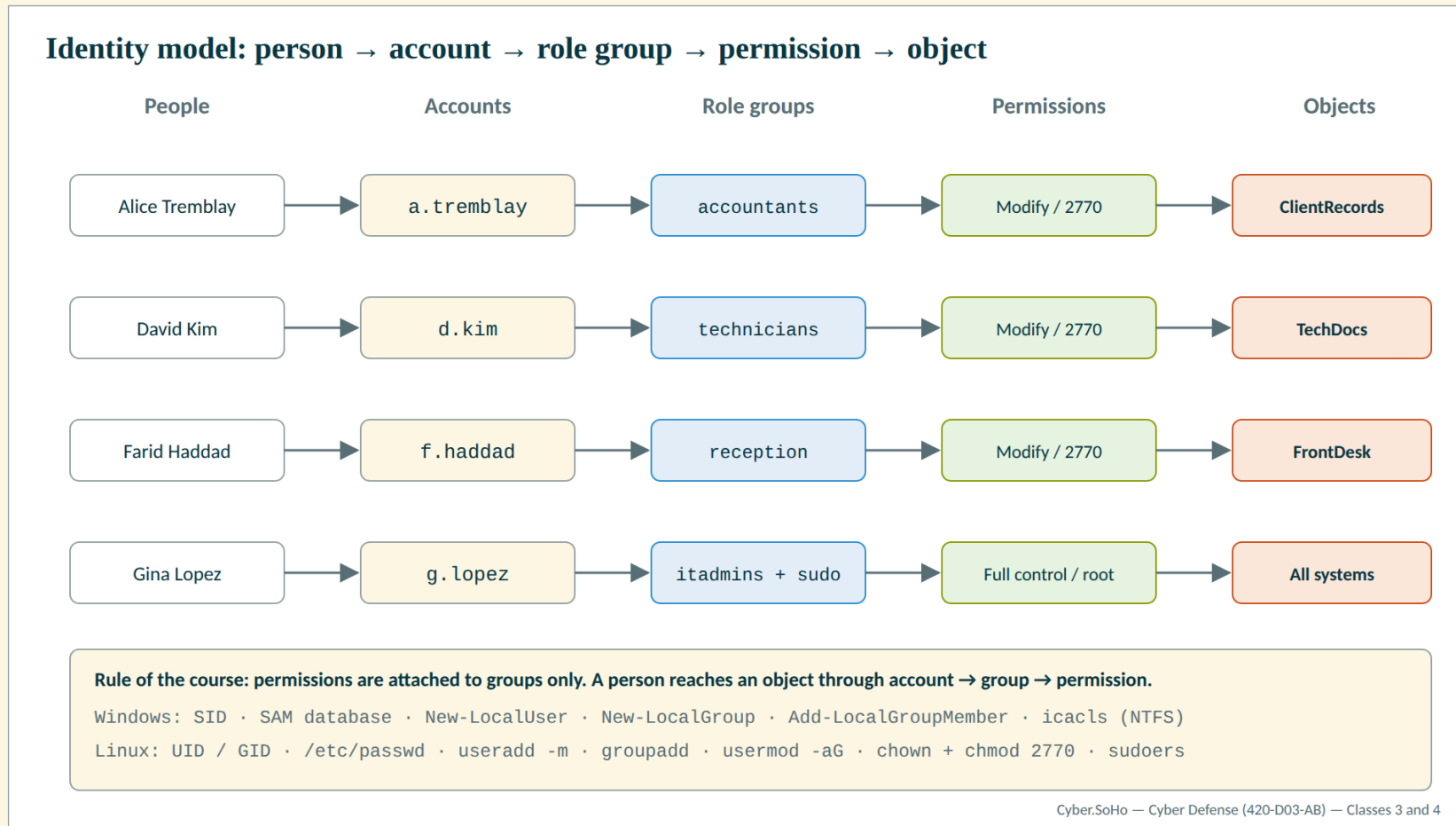


1.3 Windows and Linux, side by side (2/2)

Concept	Windows (local)	Linux
Identity store	SAM (Security Account Manager) database	/etc/passwd, /etc/shadow
Unique identifier	SID (Security Identifier)	UID (User Identifier), GID (Group Identifier)
Highest privilege	Administrator, Administrators	root (UID 0), sudo / wheel
Create a user	New-LocalUser	useradd -m
Create a group	New-LocalGroup	groupadd
Add a member	Add-LocalGroupMember	usermod -aG
List members	Get-LocalGroupMember	getent group
Force password change	net user /logonpasswordchg:yes	chage -d 0



1.3 The identity model of the course



Person → account → role group → permission → object. Permissions are attached to groups only.



1.3 Example and scripts — nm_identities.sh / nm_identities.ps1

- Four role groups: accountants, technicians, reception, itadmins.
- Seven fictional staff accounts, first-initial.lastname.
- usermod -aG: append, never replace.
- Laboratory-only initial password, expired at once with chage -d 0.
- Evidence: getent group for every role group (and Get-LocalGroupMember on Windows).
- Both scripts are accepted starting points for LAB_02 part 1 — state it in the manual.



1.4 The identity life cycle (1/2)

- Joiner: **HR** (Human Resources) confirms → owner approves → create, add to the role group, temporary password, record.
- Mover: remove the old role group before adding the new one; review what remains.
- Leaver: disable on the last day, revoke **VPN** (Virtual Private Network) and keys, rotate shared secrets, delete after retention.
- Every event ends with a record: date, requester, approver, evidence.
- Disable, not delete: files stay attributable and the audit history stays readable.

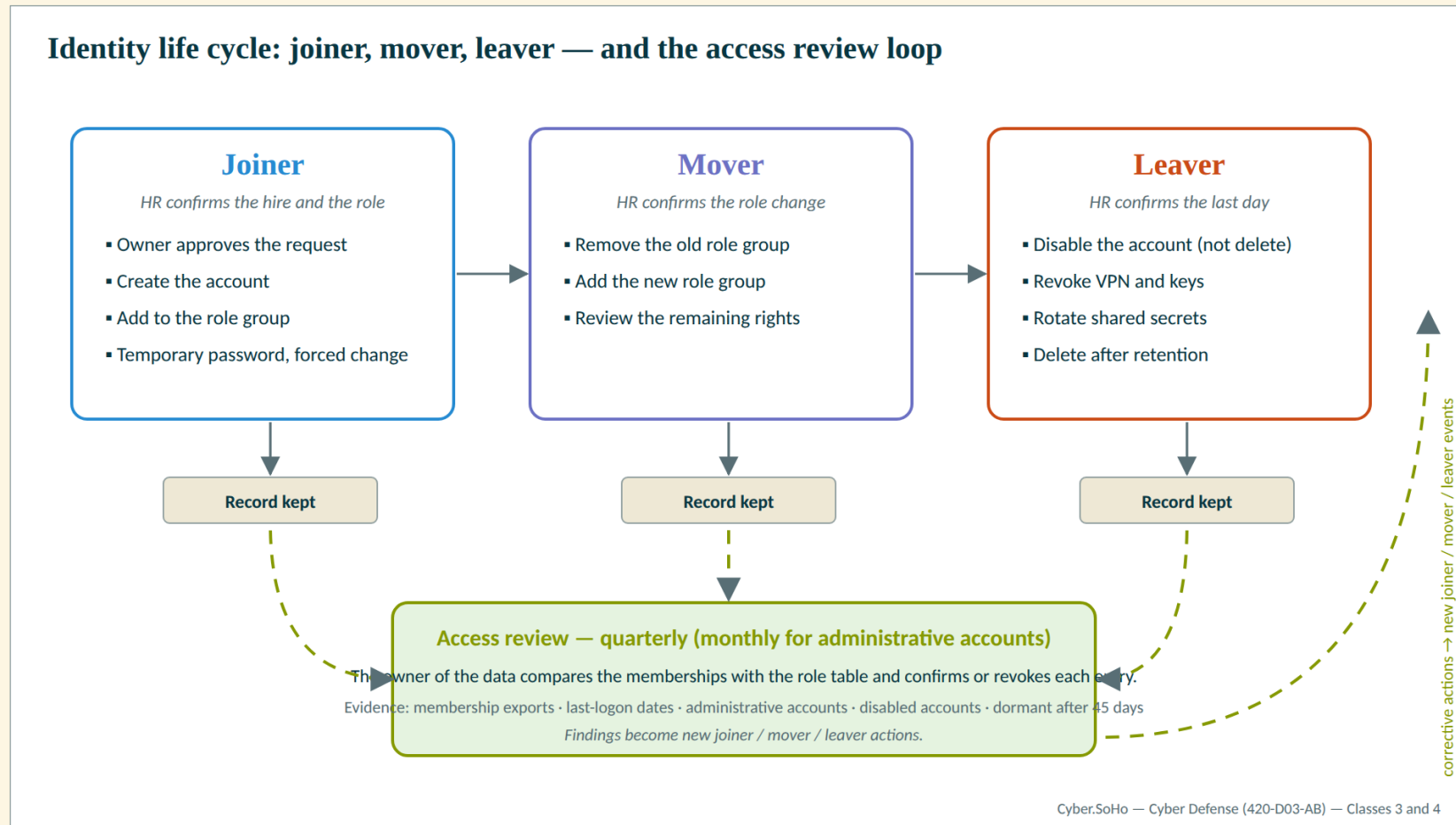


1.4 The access review (2/2)

- Quarterly for user accounts, monthly for administrative accounts.
- The owner of the data compares memberships with the role table and confirms or revokes each entry.
- Evidence set: membership exports, last-logon dates, list of administrative accounts, list of disabled accounts.
- Dormant accounts are disabled after 45 days (**CIS** (Center for Internet Security) Control 5.3); Control 6 requires the granting and revoking process.
- Tools at our scale: command-line exports and the review template; larger organizations: directory and **IAM** (Identity and Access Management) platforms.



1.4 Joiner, mover, leaver — and the review loop



Three events, one recurring control.



1.4 Example — nm_access_review.py

- Inputs: the **HR** (Human Resources) role table (authoritative) and the memberships exported from the server.
- Findings: one leaver still enabled, one mover with the old group, one joiner without account, one orphan service account.
- Actions printed: DISABLE, REMOVE + GRANT, CREATE + GRANT, DISABLE and investigate, OK.
- The actions are applied with the commands of section 1.3, then a second export proves the result.



Lesson 1 — checkpoint

- Subject, operation, object — the three parts of every decision.
- Permission on the object; privilege on the identity.
- Least privilege limits the level; need-to-know limits the scope.
- One group per role; permissions on groups only.
- Joiner, mover, leaver; review with evidence; disable before delete.
- Flash cards and quizzes for sections 1.1 to 1.4 are in the lecture document — work time after the break.



Lesson 2 — Class 4

Network Access Control and Security — part 2
— LAB_02 introduction

- 2.1 Windows: **NTFS** (New Technology File System) permissions, sharing, Active Directory
- 2.2 Linux: ownership, permission bits, sudo
- 2.3 Techniques for securing sensitive data and systems
- 2.4 Malware families, detection and prevention



2.1 NTFS permissions (1/3)

- Every file and folder carries a security descriptor: owner **SID** (Security Identifier), **DACL** (Discretionary Access Control List) (access), **SACL** (System Access Control List) (auditing).
- An **ACE** (Access Control Entry) = a **SID**, Allow or Deny, a set of rights, inheritance flags.
- Six basic permissions: Full control, Modify, Read & execute, List folder contents, Read, Write.
- Modify = read + write + execute + delete, without changing permissions or taking ownership.
- Role groups get Modify on their folder; Administrators and SYSTEM keep Full control.



2.1 The five evaluation rules and sharing (2/3)

- 1 — Rights are cumulative across all the groups of the identity.
- 2 — An explicit Deny overrides any Allow.
- 3 — An explicit entry overrides an inherited one.
- 4 — No entry at all means no access: the implicit deny.
- 5 — Inheritance flows from folder to content: (OI) files, (CI) sub-folders, unless disabled.
- Through the network (**SMB** (Server Message Block) share): effective access = the more restrictive of share permissions and **NTFS** (New Technology File System) permissions.



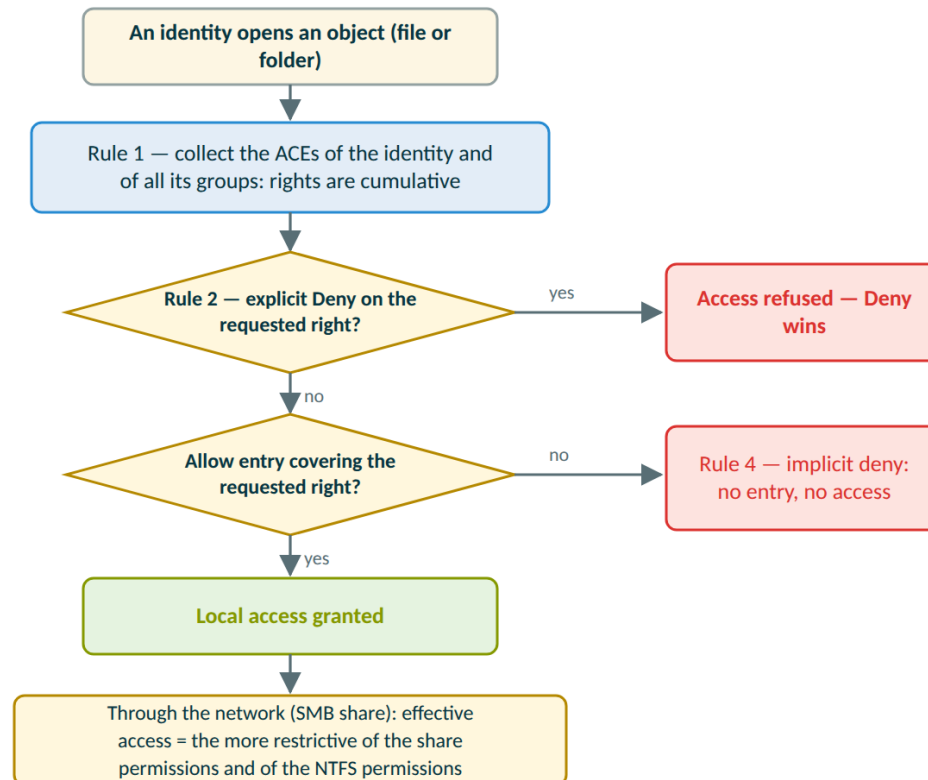
2.1 Active Directory at awareness level (3/3)

- **AD DS** (Active Directory Domain Services): a central directory on domain controllers (**DC** (Domain Controller)); domain accounts and groups valid on every member computer.
- Objects organized in **OU** (Organizational Unit) containers; a **GPO** (Group Policy Object) pushes password, lockout, user-rights and firewall settings to an **OU**.
- Authentication with Kerberos: time-limited tickets, the workstation never sees the password in clear.
- Group nesting: **AGDLP** (Accounts, Global groups, Domain Local groups, Permissions) — accounts in global groups, global groups in domain local groups, permissions on the latter.
- LAB_02 uses local accounts on standalone machines: no domain controller is required.



2.1 How Windows decides

How Windows decides: NTFS entries, inheritance, and share permissions



Rules 3 and 5 — where the entries come from

Rule 3: an explicit entry set on the object overrides an inherited entry received from the parent folder.

Rule 5: entries flow from a folder to its content unless inheritance is disabled — `icacls` flags `(OI)` for files, `(CI)` for sub-folders. `/inheritance:r` removes the inherited entries.

Test on D:\NorthMaple\ClientRecords

Administrators	(OI)(CI) Full control
Accountants	(OI)(CI) Modify
SYSTEM	(OI)(CI) Full control

a.tremblay (Accountants)	Modify — allowed
f.haddad (Reception)	no entry — refused
d.kim (Technicians)	no entry — refused

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Cumulative rights, Deny wins, explicit beats inherited, absence means no access; the share adds a second limit.



2.1 Example — Set-NorthMapleNtfs.ps1

- `icacls /inheritance:r` — remove inheritance and the inherited entries.
- `icacls /grant Administrators:(OI)(CI)F, Accountants:(OI)(CI)M, SYSTEM:(OI)(CI)F.`
- New-SmbShare ClientRecords — Change for Accountants, Full for Administrators.
- Evidence: `icacls` output and `Get-SmbShareAccess`.
- Validation by test logins: a.tremblay creates a file; f.haddad and d.kim receive Access is denied.



2.2 Ownership and permission bits (1/2)

- Every file: one owner (**UID** (User Identifier)), one group (**GID** (Group Identifier)), one mode.
- Three classes — user, group, others — with r (4), w (2), x (1).
- On a folder, x = the right to enter; r alone lists names only.
- Octal: rwx = 7, rw- = 6, r-x = 5, r-- = 4; 750, 770, 640, 600.
- Special bits as the leading digit: setuid 4, setgid 2, sticky 1 — role folders use 2770.
- One class decides: owner, otherwise group, otherwise others — no cumulation; root bypasses read and write checks.



2.2 sudo — controlled administrative access (2/2)

- Run specific commands as another user (root) after typing your own password; every use is logged.
- Rules in `/etc/sudoers` and `/etc/sudoers.d/`, edited only with `visudo` (syntax check).
- Syntax: `who where=(as_whom) commands`.
- `%technicians ALL=(root) /usr/bin/apt update, /usr/bin/apt upgrade` — least privilege for administration.
- `sudo -l` lists what you may run; `NOPASSWD: ALL` is never used for a role account.
- Root login over **SSH** (Secure Shell) is disabled in LAB_03: every action goes through a named account and `sudo`.



2.2 Anatomy of ls -l and of the octal mode

Anatomy of ls -l and of the octal mode

```
drwxrws--- 2 root accountants 4096 Sep 8 18:40 client-records
```

d = type: directory **rw**x = owner (u) **rws** = group (g), with setgid **---** = others (o)
2 = links **root** = owner **accountants** = group **client-records** = name of the folder

Bit values and the octal digit of each class

r = 4 read owner **rw**x = 4+2+1 = 7
w = 2 write group **rws** = 4+2+1 = 7 (+ setgid)
x = 1 execute / enter a folder others **---** = 0

special bits (leading digit): setuid = 4, setgid = 2, sticky = 1 → mode 2770

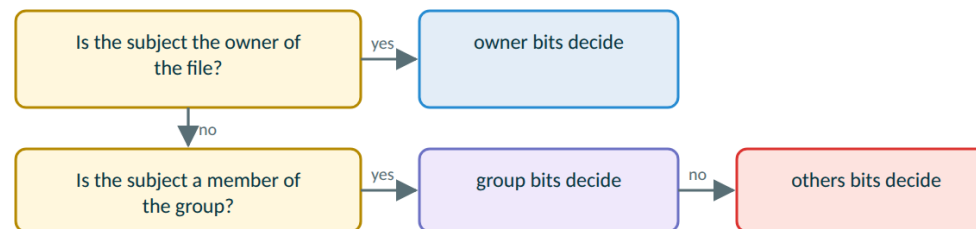
Special bits

setuid (4): a program runs with the identity of its owner — /usr/bin/passwd writes to /etc/shadow.

setgid (2): on a folder, new files inherit the group of the folder — essential for a team folder.

sticky (1): in a shared folder, only the owner of a file may delete it — /tmp is 1777.

The evaluation rule — one class only, no cumulation



root bypasses the read and write checks (execution still needs one x bit).
Exceptions for one extra user: POSIX ACLs with setfacl / getfacl.
sudo: who where=(as_who) commands — %technicians ALL=(root) /usr/bin/apt update, /usr/bin/apt upgrade; edited with visudo only.

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Type, three classes of three bits, values 4-2-1, the special bits, and the one-class evaluation rule.



2.2 Example — nm_linux_permissions.sh

- `mkdir -p`, `chown root:group`, `chmod 2770` on `client-records`, `techdocs`, `front-desk`.
- `ls -ld` as evidence: `drwxrws---` root accountants ... `client-records`.
- A drop-in `sudoers` file for technicians, mode 440, checked with `visudo -c`.
- Validation with `sudo -u`: the accountant writes (ALLOWED), the receptionist is refused (DENIED).



2.3 Securing access to sensitive data (1/2)

- Sensitive data: its loss of confidentiality, integrity or availability (**CIA** (Confidentiality, Integrity, Availability)) causes damage — client records are confidential.
- 1 — Stronger authentication: 12-character minimum, no reuse, **MFA** (Multi-Factor Authentication) with **TOTP** (Time-based One-Time Password), **SSH** (Secure Shell) keys.
- 2 — Account lockout and rate limiting: 5 attempts per 15 minutes.
- 3 — Privileged account separation: named administrative accounts, built-in Administrator disabled, no shared accounts.
- 4 — Authorization by role: groups, least privilege, need-to-know.



2.3 Securing access to sensitive data (2/2)

- 5 — Encryption at rest (BitLocker, LUKS, VeraCrypt) and in transit (**SMB** (Server Message Block) signing, **HTTPS** (Hypertext Transfer Protocol Secure), **VPN** (Virtual Private Network)) — LAB_06.
- 6 — Monitoring: **SACL** (System Access Control List) auditing, failed logons, auditd, log review, the **IDS** (Intrusion Detection System) of LAB_04.
- 7 — Availability: 3-2-1 backups — three copies, two media, one off-site — with a tested restore.
- 8 — Hygiene: screen lock, session time-outs, security updates (LAB_03), removal of unused services and accounts.
- Baseline of the Canadian Centre for Cyber Security; **CIS** (Center for Internet Security) Controls 3, 4, 5 and 6.



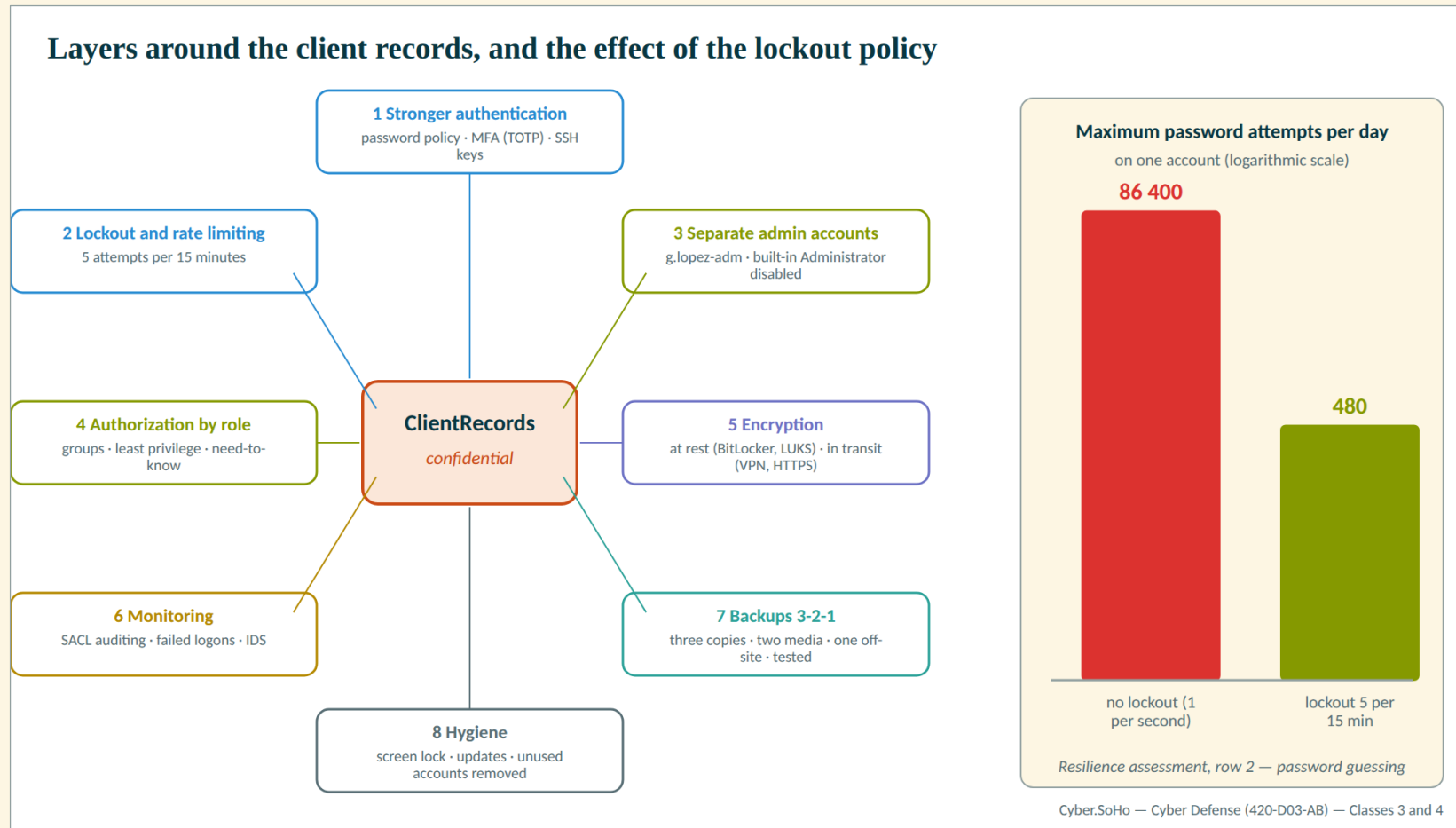
2.3 The lockout policy, as a number

86 400 → 480

- Password attempts per day on one account: one per second without lockout, 5 per 15 minutes with lockout.
- Row 2 of the resilience assessment: repeated password guessing against the remote access.
- The policy is set with net accounts on Windows, pam_faillock on Linux, fail2ban for exposed services.



2.3 Layers around the client records



Eight techniques, placed on the process each one protects.



2.3 Example — Harden-AccountAccess.ps1

- net accounts before; /minpwlen:12 /maxpwage:90 /uniquepw:5; /lockoutthreshold:5 /lockoutduration:15 /lockoutwindow:15.
- New-LocalUser g.lopez-adm; Add to Administrators; test logon; only then Remove g.lopez from Administrators.
- Disable-LocalUser Administrator.
- auditpol File System success and failure; one FileSystemAuditRule on ClientRecords (failures).
- net accounts after — the before/after pair is the LAB_03 checklist evidence.



2.4 Malware families that abuse weak access control (1/2)

Family	What it is	Weakness abused
Backdoor	Hidden access bypassing authentication	Admin rights obtained; no account or service review
Trojan	Legitimate-looking program with hidden actions	Daily account holds administrative rights
Rootkit	Hides itself by modifying the OS (Operating System)	Admin or kernel privileges; unpatched vulnerability
Worm	Self-replicates across the network	Writable shares, reused passwords, unpatched services
Bot / botnet	Remotely controlled machines (C2 (Command and Control))	Persistence through admin rights; default credentials



2.4 Detection and prevention (2/2)

Family	Basic detection	Basic prevention
Backdoor	Unexpected accounts, listening ports, scheduled tasks	Least privilege, access reviews, host firewall
Trojan	Antivirus / EDR (Endpoint Detection and Response) alerts, unusual processes and connections	No admin for daily work, application control, awareness
Rootkit	File integrity baseline, offline scan, Secure Boot	Least privilege, updates, signed drivers
Worm	IDS (Intrusion Detection System) alerts, traffic spikes, failed logons	Updates, zoning, host firewalls, least privilege on shares
Bot / botnet	Periodic outbound connections to unknown hosts	Changed default credentials, outbound filtering, updates



2.4 The weakness each family abuses, and the layer that answers

Malware families: the weakness each one abuses and the layer that answers			
Family	Weakness of access control abused	Basic detection	Defense layer — lab of the course
Backdoor	Administrative rights obtained by the attacker; no review of accounts and services.	Unexpected accounts or members; listening ports (ss -tlnp); scheduled tasks and startup entries.	Least privilege and access review (LAB_02); host firewall (LAB_03); integrity baseline.
Trojan	The user runs it with a daily account that holds administrative rights.	Antivirus and EDR alerts; unusual processes; unexpected outbound connections.	No admin for daily work (LAB_02); application control; attachment filtering; awareness.
Rootkit	Administrative or kernel privileges through weak separation or an unpatched vulnerability.	File integrity baseline (hashes); offline scan; rkhunter, chkrootkit; Secure Boot.	Least privilege (LAB_02); security updates (LAB_03); signed drivers; Secure Boot.
Worm	Shares writable by Everyone; reused administrative passwords; unpatched services reachable across zones.	IDS alerts (LAB_04); traffic spikes between machines; many failed logons.	Updates (LAB_03); zoning (LAB_01, LAB_05); host firewalls (LAB_03); least privilege on shares (LAB_02).
Bot / botnet	Persistence through administrative rights; default credentials on network devices (row 9).	Periodic outbound connections to unknown hosts in firewall and IDS logs; unexpected boot-time processes.	Changed default credentials (LAB_05); outbound filtering; updates; least privilege.

Ransomware combines Trojan delivery, worm-like spread and encryption of what the account can write to (row 10 of the resilience assessment).
Prevention in order of effect: least privilege, security updates, host firewalls, changed default credentials, tested backups.

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Every row maps to a lab of the course.



2.4 Example — nm_integrity_check.py

- init: hash six critical files with **SHA-256** (Secure Hash Algorithm, 256-bit) and store the baseline on a read-only medium.
- check: recompute and report OK, CHANGED or MISSING per file.
- CHANGED on /usr/bin/ps: rootkit indicator. CHANGED on /etc/crontab: backdoor persistence.
- Response: isolate, restore from a clean snapshot, rotate credentials, document (Chapter 7).
- Class 14 practises hashing; LAB_03 uses the same technique for before/after evidence.



Lesson 2 — checkpoint

- **NTFS** (New Technology File System): cumulative, Deny wins, explicit beats inherited, absence means no access, inheritance propagates; share limits network access.
- Linux: owner, group, others; 4-2-1; one class decides; 2770 for role folders; sudo rules with visudo.
- Eight techniques, each attached to the process it protects; 86 400 becomes 480.
- Six families, each abusing a weakness we removed today; detection tells you the layers are being tested.
- Flash cards and quizzes for sections 2.1 to 2.4 are in the lecture document.



LAB_02 — Privilege and Access Management (1/2)

- Objective: implement the role table on one Windows and one Linux machine of your isolated laboratory, and prove it with test logins.
- Part 1 (this session): role groups and staff accounts on both machines; least-privilege memberships; test logins with screenshots.
- Part 2: role folders with **NTFS** (New Technology File System) and share permissions (Windows), ownership and mode 2770 (Linux); validation checklist.
- Deliverable: permission tables and validation screenshots — Chapter 2 of the Final Project and of the manual.
- Scripts of this lecture are accepted starting points; state the reuse in the manual.



LAB_02 — rules and evidence (2/2)

- Isolated laboratory only; snapshots before each part.
- Laboratory-only accounts and passwords; nothing real in any screenshot.
- Evidence: access matrix, membership exports, permission tables, allowed and refused test logins per role.
- Equivalent open-source tools and your own helper scripts are welcome — stated in the manual.
- **AI** (Artificial Intelligence) tools allowed with disclosure: the **LLM** (Large Language Model) used and the exact prompts, verbatim; local models (Ollama, Jan, Open WebUI, GPT4All, LocalAI) run inside the laboratory.
- Submission on Omnivox Léa DropBox, 420-D03-AB_C04_LAB02_LASTNAME_Firstname; deadline in the LAB_02 document.



Links used today (1/2)

- **CIS** (Center for Internet Security) Critical Security Controls — [cisecurity.org/controls](https://www.cisecurity.org/controls) (Controls 5 and 6).
- **NIST** (National Institute of Standards and Technology) CSRC glossary — csrc.nist.gov/glossary (least privilege).
- Canadian Centre for Cyber Security — cyber.gc.ca (baseline controls, top 10 actions).
- Microsoft Learn — Access Control Lists, `icacls`, LocalAccounts module, **AD DS** (Active Directory Domain Services) overview.
- man7.org — `useradd(8)`, `chmod(1)`, `chown(1)`, `sudoers(5)`; sudo.ws.



Links used today (2/2)

- MITRE **ATT&CK** (Adversarial Tactics, Techniques, and Common Knowledge) — attack.mitre.org (T1078, T1014, T1110).
- **CISA** (Cybersecurity and Infrastructure Security Agency) — cisa.gov; ClamAV — clamav.net; Microsoft Defender Antivirus.
- FreeIPA — freeipa.org; Keycloak — keycloak.org (awareness).
- Criminal Code of Canada, section 342.1 — laws-lois.justice.gc.ca.
- VirusTotal **URL** (Uniform Resource Locator) scanner — virustotal.com/gui/home/url.
- Local **AI** (Artificial Intelligence) tools: ollama.com, jan.ai, openwebui.com, gpt4all.io, localai.io.



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- Full text: Annex of the lecture document; session signature at the beginning of each session.





Next: work time — LAB_02 part 1

After the break: flash cards and quizzes of both lessons, LAB_02 part 1, Chapter 2 of the manual started this week.

Next: Class 5 — Wednesday, September 9 — Securing Windows and Linux Hosts — LAB_03 (start)

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